Salient Is Not Enough: Keeping Every Time Anchor for Temporal Grounding under a Visual-Token Budget

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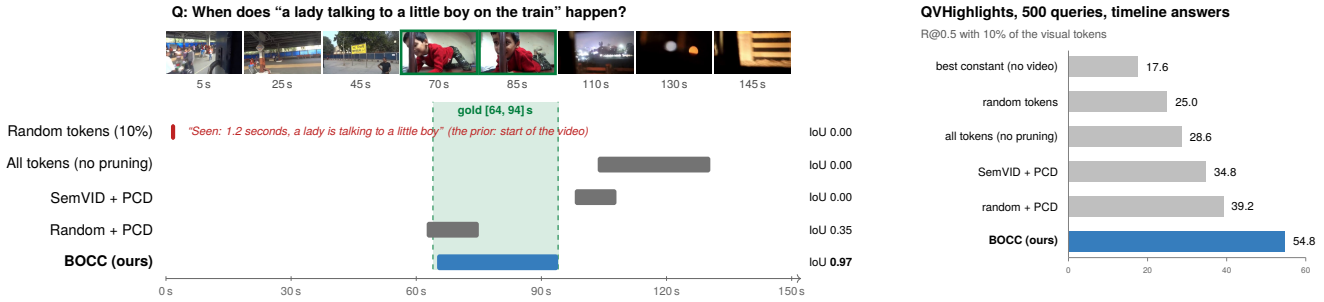


Figure 1. **Temporal grounding from a pruned video memory.** Left: a QVHighlights query, with the intervals predicted by each method at 10% of the visual tokens (real outputs). With random pruning the model falls back on a video-free prior and answers "1.2 seconds"; QA-oriented pruning and the uncompressed model pick a later, wrong scene. BOCC keeps every frame anchored in time and contrasts the answer against the model's no-video prediction, and recovers the interval (IoU 0.97). Right: R@0.5 over 500 queries when every method answers with a timeline of observations, the format of grounded QA. In this format the prior writes many of the times and BOCC is far above the uncompressed model and the best video-free constant. With a one-line answer the uncompressed model is much stronger, and Coverage alone matches it with a quarter of the tokens (Sec. 6.2).

Abstract

Video LLMs are increasingly asked not only what *happened but* when*: a grounded question such as "how long did I use the paper cutter?" needs an answer and the time interval of its evidence. To keep long videos tractable, the visual memory of these models is pruned to a small fraction of its tokens, and published pruners lose little multiple-choice accuracy at 10% of the tokens. We show that temporal grounding needs something different:* pruning for QA keeps what is salient, but grounding needs what is everywhere. *A video LLM reads time from timestamps that survive only together with visual tokens of their frame, so pruners that select by saliency, uniqueness or text similarity empty frames, delete their timestamps, and fall* below random token selection *on grounding. When the answer lists observed moments before the interval, the gaps are filled by a video-free prior: asked with no video at all, the model still starts its intervals at the beginning of the window. We propose BOCC, a training-free method that selects frames by the question (BOLT) and adds two parts.* Coverage *gives every frame an equal token quota, chosen per layer by the actual question's attention, so that every time anchor survives.* Prior-contrastive decoding *(PCD) writes each time value against the model's own no-video prediction. The two parts behave differently across answer formats, which we study directly. Under the standard one-line grounding prompt, Coverage alone matches the uncompressed model with a quarter of the visual tokens on a general and a grounding-tuned model of the same family (QVHighlights, 61.7* vs. *60.5 and 67.3* vs. *67.1 mIoU) and beats random selection (+3.5 and +2.0), while PCD has no effect; on the full Charades-STA and QVHighlights sets of TimeLens-Bench, Coverage at 31% of the tokens is lossless on both models and beats both random selection and a lower-resolution input of the same budget. In reasoning-style grounded QA (S-EMBER), PCD is what matters: with 10% of the tokens BOCC improves on the uncompressed model by +11.2 and +10.7 mIoU under the same prompt, and matches an oracle that sees only the gold evidence on grounded answers. Asking for a timeline of observations instead of a single interval costs single-moment grounding about 30 mIoU, so the answer format is a first-order variable in evaluating compressed memories. We evaluate every method against video-free floors and re-*

port where the method helps less: on short, densely sampled videos (Charades-STA, NExT-GQA) the gain of Coverage is small at moderate budgets and grows as the budget shrinks (+6.8 accuracy over random selection at 5% on NExT-GQA), and PCD helps only where the answer format lets the prior choose the times.

1. Introduction

Ask a video assistant "how long did I use the paper cutter?" and a useful answer has two parts: a duration and the interval it comes from. Such *grounded* questions are central to egocentric assistants, video search and step-by-step guidance. They are hard for the same reason they are useful: the model must know *when* things happened across the whole window, including the moments just before and after the event.

At the same time, video LLMs pay for every frame. Qwen3-VL [38] spends about 330 visual tokens per frame (660 per pair of frames), so 64 frames already cost about 21,000 tokens, and long or streaming video multiplies this. Pruning the visual tokens to a small budget is therefore the default, and a fast-growing literature reports that 10–25% of the tokens preserve multiple-choice accuracy [14, 18, 27, 35, 41, 44]. This paper asks what pruning does to *temporal grounding*. Our answer is a single principle with two consequences: **pruning for QA keeps what is salient, but grounding needs what is everywhere.** A QA answer can be read from a few informative frames; an interval can only be read from a timeline, and a timeline needs an anchor in every part of the window.

Observation 1: QA pruners lose the anchors. Video LLMs such as Qwen3-VL report time by reading timestamp tokens interleaved with the visual tokens. A frame whose visual tokens are all removed loses its timestamp, or keeps a timestamp with nothing to show. Pruners designed for QA remove frames that look redundant, unremarkable or unrelated to the words of the question, and those frames are exactly what makes a timeline usable. On S-EMBER grounded QA, all four recent pruners we evaluate fall below *random* token selection at the same budget, and the same holds on QVHighlights and Charades-STA (Sec. 6.3). Random selection is a strong baseline for grounding because it keeps a few tokens on every frame, and one of the published scores improves by 5.3^{*} mIoU simply by being ranked within each frame instead of globally (Sec. 3.2).

Observation 2: where the memory is silent, a prior answers. Grounded QA invites a reasoning-style answer: the model lists what it saw and when, then states the interval. Asked this way with no video at all, the model still produces confident intervals, and 90% of them start in the first 5% of the window. A pruned memory shows the same habit: 30–56% of predicted intervals start in the first 5% of the window, against 16% of the gold intervals, and removing the frames that contain the evidence raises this share from 37% to 51% (Sec. 3.3). The prior is therefore not a property of a particular pruner; it is what the decoder writes where the memory has nothing to say.

Observation 3: the answer format decides which failure dominates. The two observations call for two remedies, and they do not matter equally everywhere. On the same QVHighlights queries and the same memory, asking for a timeline of observations instead of a single interval costs about 30 mIoU (Sec. 6.2). Under the one-line format, keeping anchors is what helps and contrastive decoding does nothing; under the timeline format, the video-free prior writes many of the times, and contrasting it away is what helps. We therefore treat the answer format as an experimental variable, not a fixed detail of the protocol.

Observation 4: benchmark floors are high. Grounding benchmarks have strong positional regularities [37]. On S-EMBER, where questions are asked mid-stream and two thirds of the gold intervals end within the last fifth of the window, the best *constant* interval $[0.15q, q]$ reaches 36.9 mIoU, more than the uncompressed model answering in the timeline format (28.0). Following Otani et al. [37], we report video-free floors for every method (Sec. 3.4).

We propose BOCC (BOLT + Coverage + Contrastive decoding), a training-free method with one component for each failure, behind an optional BOLT front end that chooses which frames to encode (Fig. 3):

- **Coverage** (Sec. 4.1) keeps the anchors. After the frames are encoded, the KV cache is pruned layer by layer. Every frame keeps an equal share of the layer's budget, and within each frame the tokens that the *actual question* attends to most are kept. Because a timestamp survives exactly when one of its frame's visual tokens survives, the equal quota keeps every time anchor, and the question decides what each anchor shows.
- **Prior-contrastive decoding (PCD)** (Sec. 4.2) removes the prior where it writes the answer. Each time value is chosen by contrasting the model's prediction with its prediction from a video-free memory, the content-free contrast of contextual calibration and context-aware decoding [42, 52] applied only to time values, with a strength that fades where the model is already confident.

Neither operator is new in isolation: per-frame budgets appear in VidCom^2, KTV and AdaTP [1, 12, 35], question-attention KV eviction in SnapKV and PruneVid [27, 32], and contrast against a content-free or degraded input in contrastive decoding, classifier-free guidance and VCD [19, 30, 31, 40, 42]. Our contributions are to show *why* temporal grounding needs each of them, *when* each one matters, and to make them work together:

1. **A diagnosis of grounding under compression** (Sec. 3).

QA-oriented pruners empty frames and fall below random selection; in reasoning-style answers the decoder's video-free prior fills the gaps, which an evidence-removal test shows causally.

2. **Coverage**, a per-layer equal quota filled by the question (Sec. 4.1). Under the standard one-line prompt it matches the uncompressed model with 24% of the visual tokens on Qwen3-VL-8B and the grounding-tuned TimeLens-8B, beats random selection, SemVID and a matched low-resolution input, and at 897 tokens exceeds a 1 fps input of 1,125 tokens by $+8.8^*$ mIoU; on the full Charades-STA set it matches the uncompressed model and beats random selection (Sec. 6.2).
3. **PCD and an answer-format study** (Secs. 4.2 and 6.2). PCD helps reasoning-style answers on all eight memories we test and on both models; with 10% of the tokens BOCC improves on the uncompressed model by $+11.2^*$ and $+10.7^*$ mIoU on S-EMBER grounded QA under the same prompt. The same study shows that the answer format alone moves single-moment grounding by about 30 mIoU and decides whether contrastive decoding helps.
4. **Floor-adjusted, reproducible evaluation** (Sec. 5). Video-free floors and a length-rescaling baseline for every method, paired bootstrap tests, a reproduction of the published TimeLens-Bench numbers within 0.3 mIoU, and a broad set of negative results, including the limits of Coverage on short, densely sampled videos.

2. Related Work

Frame selection. BOLT [34] scores frames against the query with a contrastive image–text model and selects them by inverse transform sampling; AKS [43], MDP3 [5], FOCUS [2] and Q-Frame [6] optimise relevance together with coverage or diversity. All are evaluated on multiple-choice QA. Our frame selection front end (Sec. 4.3) adapts BOLT to grounding, and we find that it contributes little once tokens are chosen by the question.

Visual token pruning for video LLMs. Query-agnostic methods compress by redundancy or saliency: FastVID [41] segments the video and keeps density peaks, VidCom^2 [35] gives each frame a budget by its uniqueness, FlashVID [18] selects by [CLS] attention and merges tokens across frames with trees, and DyCoke [44] and FrameFusion [21] merge similar tokens. Query-aware methods use the text: SemVID [14] scores tokens by word-embedding relevance with MMR diversity, AdaTP [1] boosts segments similar to the text, LGTTP [4] reads temporal words in the question to reallocate tokens, and PruneVid [27] keeps the top 40% of tokens by question attention at one LLM layer. KTV [12] is closest in structure (keyframes, then tokens with a relevance-weighted per-frame budget), and LGTTP is the only method we know that evaluates grounding, at a milder budget and offline. BOCC differs in three ways: the per-frame quota is equal by design, the score is the actual question's attention propagated exactly through all layers, and the quota exists to keep every timestamp, which we show matters for grounding.

KV-cache compression and video memory. SnapKV [32], H_2O [51], PyramidKV [15] and AdaKV [20] evict KV entries by attention in long-context LLMs; our token stage is SnapKV-style eviction applied to video with a per-frame constraint. For streaming video, HERMES [10] compresses the cache with generic pseudo-queries, ReKV [17] retrieves KV blocks at question time, and StreamMem [8] and InfiniPot-V [3] maintain bounded memories. These works report QA only. Our memory is built per question from the frames up to the question time: it is causal, but it is not a persistent streaming memory, and we make no claim about the cost of building it (Sec. 7).

Contrastive decoding and prior correction. Contrasting a model's prediction with a weaker or less informed one is an established decoding tool. Contrastive Decoding [31] subtracts a small "amateur" model and introduced the plausibility constraint we use; classifier-free guidance for language models [40] and context-aware decoding [42] amplify the difference between the predictions with and without the conditioning context, which is the form of our Eq. (4). The idea of measuring the prior with a content-free input and correcting for it is older still: contextual calibration [52] rescales label probabilities by the model's answer to "N/A", and domain-conditional PMI [24] reweighs answers by their task-conditioned prior. For vision-language models, VCD [30] contrasts against a noised image, ICD [46] against a disturbed instruction, M3ID [19] against the absence of the image, and PAI [33] amplifies image attention while subtracting text-only logits; SEASON [7] extends the idea to video with temporally corrupted negatives. These methods target hallucination, faithfulness or classification. PCD uses the same contrast for a different purpose: it is applied only to *time values*, with a confidence-adaptive strength, and we study when it helps and when it does not (Secs. 6.2 and 6.5), including whether its benefit is more than a uniform lengthening of the interval (Sec. 6.5).

Grounding-tuned video LLMs. TimeChat [39], VTimeLLM [25] and LITA [26] train video LLMs to localise moments, with timestamp-aware encoders, boundary-aware training stages or dedicated time tokens; TimeLens [50] re-annotates the standard grounding benchmarks and fine-tunes Qwen-VL models with reinforcement learning. These works improve grounding with full visual memories. We use TimeLens-8B as a grounding-tuned model and TimeLens-Bench as the standard protocol, and ask what happens to such a model when its memory is pruned.

Training-free temporal grounding. NumPro [48] draws frame numbers on the frames, FV-Action [9] scans the video with yes/no questions, REZE [13] recasts grounding as recognition, and [Authors to verify] [11] read intervals from calibrated attention heads, debiased against a blank-video run. All assume the full set of video tokens. BOCC addresses the complementary problem of grounding from a *compressed* memory; these readouts could run on top of it **[TODO: run FV-Action and the attention readout on the same 10% memory]**.

Benchmark biases. Otani et al. [37] showed that blind baselines are competitive on Charades-STA and ActivityNet Captions because of positional priors in the annotations. We find the same on the datasets we use, adopt their blind baselines as video-free floors in every comparison, and add a length-rescaling floor for methods that lengthen intervals.

3. Grounding under a Token Budget: a Diagnosis

This section establishes Observations 1, 2 and 4 of the introduction on S-EMBER, before any method is introduced; Observation 3, on the answer format, is studied in Sec. 6.2. Unless stated otherwise, numbers in this section use random 10% pruning as the reference memory.

3.1. Setting

A question x is asked at time q about a video $v_{0:q}$. The model sees F=64 frames from $[0, q]$, encoded into a KV cache $\mathcal{M}$ of N visual tokens interleaved with timestamp tokens (one "`<t seconds>`" per pair of frames in Qwen3-VL). A budget ρ (default ρ=0.1) limits the number of tokens kept in each layer to ρN, with kept timestamps counted inside the budget. The model answers with a short *timeline*:

```
Seen: t₁ seconds, …
Seen: tₖ seconds, …
Answer: …
Time: [s, e]
```

from which the interval $[s, e]$ is parsed. This reasoning-style format is natural for grounded QA, where the answer explains what was seen before it states the interval, and the whole diagnosis uses it. Section 6.2 compares it with the one-line format of grounding benchmarks (*"The event happens in s–e seconds"*). Grounding is scored by its IoU with the gold interval (mIoU and R@$\{0.3, 0.5, 0.7\}$); grounded QA additionally requires the answer to be judged correct (GQ@0.5: correct answer and IoU $\geq$ 0.5). We call an answer an *early start* when its interval starts in the first 5% of the window, and a *leak* when it starts early although the gold evidence starts later.

Table 1. **Where pruning keeps tokens.** Share of the visual tokens kept, by position in the window (S-EMBER, 10% budget), and the resulting grounding (mIoU) and leak. FastVID (mIoU 17.9, leak 38%) keeps 6.0 / 8.5 / 10.5 / 10.6 / 12.9% over the five fifths of the window, and tokens from only 25% of the frames.

Selector	0–10%	10–25%	25–50%	50–75%	75–100%	mIoU	Leak
HERMES	4.7	3.9	2.6	2.5	27.6	18.3	49%
Stratified	8.7	8.7	8.6	8.6	8.7	<u>23.7</u>	**29%**
Random	8.9	8.8	8.6	8.7	8.6	**25.8**	<u>30%</u>

3.2. QA pruners lose temporal coverage

At 10% of the tokens, FastVID, VidCom^2, FlashVID and SemVID all score below random token selection (Sec. 6.3). The cause is visible in where the kept tokens lie (Tab. 1). HERMES's attention-plus-recency score keeps 27.6% of the tokens in the last quarter of the window and only 2.5–4.7% elsewhere; FastVID keeps a token from only 25% of the sampled frames in a median question, because it retains one representative per run of similar frames, and runs of similar frames are exactly where a continuing event lives. Random selection keeps about 9% everywhere and never empties a frame. A timestamp is useful only when visual content of its frame survives with it, so a selector that empties frames leaves holes in the timeline. Holes hurt even when the score itself is reasonable: ranking HERMES's scores within each frame (an equal per-frame share, "stratified") instead of globally gains 5.3* mIoU.

3.3. The prior fills the gaps

The model answers without video. Asked every question with an empty visual memory, the model still writes a timeline: 89% of its first `Seen:` moments are at 0 s and 90% of its intervals start in the first 5% of the window (88% leak). Its intervals are 0.45× the gold length (median) and 87% of its ends are more than half an interval too early. We call this the model's *temporal prior*: start early, stop soon.

The prior leaks into answers with video. The same signature appears in every memory, weaker (Fig. 2). About 30% of intervals start early at every random budget and with no pruning at all, against 16% of gold intervals, and the share rises when coverage is lost (HERMES 49%, HERMES streaming 37–39%). The leak is therefore not a matter of the number of tokens, but of where the memory is silent.

Removing the evidence causes the leak. To test causality we drop the uniform frames that fall inside the gold interval ("evidence") or the same number of frames elsewhere ("control"), keeping every other frame with its true timestamp (Tab. 2). With the evidence removed, the answers do not spread out: they move to the start of the video, and the leak rises by 14 points for both the unpruned and the random 10% memory. Within the HERMES 10% memory, ques-

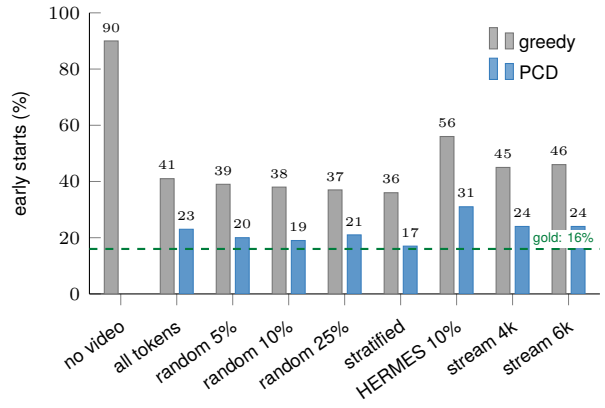


Figure 2. **The temporal prior leaks into every memory, and PCD removes most of it.** Share of predicted intervals that start in the first 5% of the window (S-EMBER). Without video the model starts early 90% of the time; every memory inherits part of this habit, most where pruning loses coverage (HERMES). PCD roughly halves it everywhere.

Table 2. **Causal test.** Frames inside the gold evidence ("evidence") or as many frames elsewhere ("control") are removed. Acc. / mIoU / GQ@0.5 and leak; leak measured on the 397 questions whose evidence starts after the first 5% of the window. PCD: Sec. 4.2.

		Greedy		PCD	
Memory	Removed	Acc./mIoU/GQ	Leak	Acc./mIoU/GQ	Leak
Random 10%	control	16.6/28.8/8.8	**31%**	15.4/34.6/10.5	**11%**
	evidence	7.8/ 8.6/1.7	46%	9.3/10.3/1.1	23%
All tokens	control	**19.6/29.4/10.1**	37%	**18.1/35.9/12.6**	15%
	evidence	9.1/ 9.4/1.9	51%	9.1/10.5/1.5	22%

tions whose evidence keeps under 5% of its tokens leak 56% of the time, against 33% when at least 15% survives.

An instruction does not remove it. Adding "The relevant moments may occur at any point in the video; do not assume they are near the start" to the prompt leaves the leak almost unchanged (29.5% → 28.0%) and grounding unchanged (−0.1 mIoU). Within this answer format, the prior acts in the output distribution and has to be removed there. Whether the same prior appears when the model answers with a single interval is an open question: there PCD has no effect (Sec. 6.2), which suggests that the prior decides fewer of the written times. **[TODO: no-video answers and leak under the one-line and direct-answer prompts.]**

The end is the weak boundary. Across memories, the median end error is 0.40–0.69 of the gold length without contrastive decoding, against 0.14–0.30 for the start, and the most frequent failure is an interval entirely *before* the gold one (33% of all answers, Tab. 25).

Table 3. **Video-free floors.** Best constant interval $[aq, bq]$ on a 20×20 grid, tuned on the test questions (each metric uses its own best constant; $[a, b]$ is the mIoU optimum), against the uncompressed model. Closed: share of questions with gold end $< 0.8q$.

	Best constant					All tokens	
Dataset	$[a, b]$	mIoU	R@0.5	R@0.7	Closed	mIoU	R@0.5
S-EMBER	$[.15, 1]$	36.9	34.7	24.2	33%	28.0	26.7
QVHighlights	$[0, 1]$	26.8	17.6	8.8	62%	29.1	28.6
ActivityNet	$[0, .95]$	37.9	35.8	20.8	56%	22.9	18.2
Charades-STA	$[0, .45]$	29.7	29.6	15.4	70%	16.6	14.2

3.4. Benchmark floors

Positional regularities make some intervals predictable without the video (Tab. 3). On S-EMBER, questions are asked while the stream is running and 67% of gold intervals end within the last 20% of the window, so the constant $[0.15q, q]$ scores 36.9 mIoU. On ActivityNet Captions, whose moments cover 38% of the video on average, $[0, 0.95q]$ scores 37.9; on Charades-STA, $[0, 0.45q]$ scores 29.7. QVHighlights is the least biased (26.8 mIoU, 17.6 R@0.5). These constants are tuned on the test questions, so they upper-bound any video-free rule of this form. On three of the four datasets the best constant beats the uncompressed model answering with a timeline (with a one-line answer, Qwen3-VL-8B is well above it on ActivityNet and Charades-STA, Sec. 6.2), which makes the floors a necessary reference for every comparison that follows.

4. Method: BOCC

BOCC acts at the two places where Sec. 3 locates the failures: the memory (Coverage) and the decoder (contrastive decoding, PCD), behind a frame-selection front end (BOLT). None of them needs training. Figure 3 gives an overview.

4.1. Coverage: equal quotas filled by the question

After the F frames are encoded, the memory is pruned once, independently in each of the L layers. Let $\mathcal{T}_f$ be the visual tokens of frame pair $f \in \{1, \dots, F'\}$ and B the per-layer budget for visual tokens.

Question relevance. The question x is appended to the full, uncompressed memory and propagated exactly through all layers, so that the queries at layer ℓ come from the real hidden states. The relevance of visual token i at layer ℓ is the attention it receives from the question, averaged over heads h and question positions j:

$$a_\ell(i) \;=\; \frac{1}{H\,|x|}\sum_{h=1}^{H}\sum_{j\in x}\mathrm{softmax}_i\Big(\tfrac{1}{\sqrt{d}}\,Q_j^{(\ell,h)}\,K_i^{(\ell,h)\top}\Big). \tag{1}$$

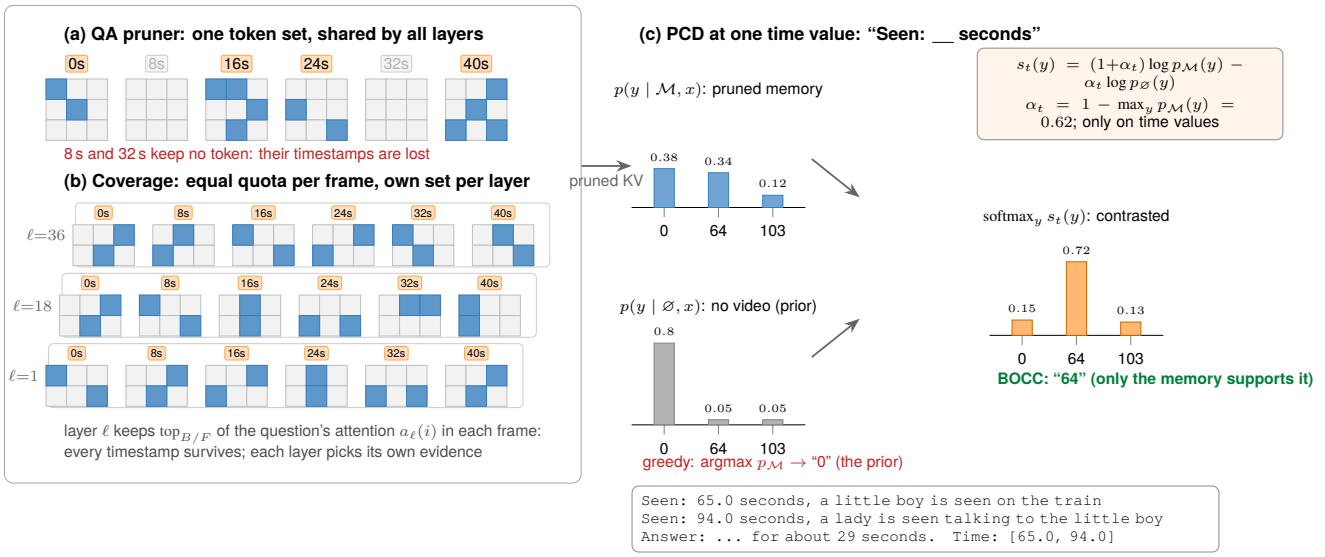


Figure 3. **Overview of BOCC. (a)** QA-oriented pruners keep one set of salient or unique tokens in every layer and can empty whole frames, removing their timestamps. **(b)** Coverage prunes each layer separately: every frame keeps the same share of the layer budget, filled with the tokens that the question attends to most *in that layer*. Every timestamp survives in every layer, and different layers keep different evidence (sharing one set across layers costs 1.5 mIoU, Tab. 16). **(c)** PCD at one time value of the teaser example (probabilities illustrative). Greedy decoding picks "0", which the memory supports only as much as the video-free prior does; the contrasted score discounts it and promotes "64", which only the memory supports. Output shown is the real BOCC answer.

Equal quota. Every frame pair receives the same share of the budget, $B_f = \lfloor B/F' \rfloor$ (remainders assigned in order), and layer ℓ keeps

$$\mathcal{K}_\ell = \bigcup_{f=1}^{F'} \operatorname{top}_{B_f} \{ a_\ell(i) \ : \ i \in \mathcal{T}_f \}. \tag{2}$$

The timestamp tokens of f are kept if and only if $\mathcal{K}_\ell \cap \mathcal{T}_f \neq \emptyset$. With an equal quota every frame pair keeps at least one token at any practical budget, so *every time anchor survives*, and the question decides which part of each frame is shown. Kept tokens retain their original rotary positions, so their temporal order is preserved.

Why equal, and why the question. The equal quota is a deliberately simple coverage constraint. It guarantees the timeline, which grounding needs and multiple-choice QA does not, and it stops attention from collapsing onto a few frames: when quotas follow attention mass, attention sinks and recency bias give a single frame up to 227 of about 2,140 kept tokens. Relevance-weighted quotas with a floor, from SigLIP similarity or attention mass, did not beat equal quotas in a pre-registered test (-0.2 mIoU). Using the *actual* question, rather than the generic probe questions of streaming memories such as HERMES [10], is what makes attention informative: generic probes with a global top-k are the weakest variant we measured (18.3 mIoU).

Per-layer selection. Each layer keeps its own set $\mathcal{K}_\ell$. Forcing one shared set (attention averaged over layers) costs BOCC -1.5^* mIoU, whereas it does not change random selection ($+0.7$, n.s.). The benefit of per-layer selection is therefore specific to question-driven scores, and it does not explain why random selection beats input-level pruners.

Frames and budget. An equal quota needs enough tokens per frame to show anything: with 2–4 tokens per frame pair, the top-attended tokens carry little content and Coverage collapses (Sec. 6.7). When the budget is small, we therefore spend it on fewer frames: at a fixed number of kept tokens, 32 frames with twice the share per frame recover what 64 frames lose. As a rule we use F=64 when each frame pair keeps at least ∼8 tokens and halve F otherwise. **[TODO: this rule was derived from the results of Sec. 6.7; validate it on held-out data.]**

Cost. Coverage adds one forward pass of the question over the uncompressed memory per question; the full memory is prefilled first, so Coverage reduces the memory the decoder reads, not the cost of building it. PCD decodes a second, prompt-only row in lockstep at every step (its scores are used only inside time values), so it roughly doubles the decoder's per-step work on a much shorter context. **[TODO: measured prefill, time to first token, total latency, peak memory and KV size; the current implementation runs 3–4 scoring passes, reducible to one.]**

4.2. PCD: prior-contrastive decoding of time values

Let $p_{\mathcal{M}}(y) = p(y \mid \mathcal{M}, x, y_{<t})$ be the next-token distribution with the compressed memory and $p_{\varnothing}(y) = p(y \mid$

$\varnothing, x, y_{<t}$) the distribution with the same prompt and generated prefix but *no visual tokens*. Then

$$\log p_{\mathcal{M}}(y) \;=\; \log p_{\varnothing}(y) \;+\; \underbrace{\log \frac{p_{\mathcal{M}}(y)}{p_{\varnothing}(y)}}_{\text{evidence from } \mathcal{M}}, \qquad (3)$$

so when the memory carries no information about y, the evidence term is flat and the video-free prior decides: this is the failure of Sec. 3.3. Prior-contrastive decoding (PCD) amplifies the evidence term,

$$\begin{aligned} s_t(y) \;&=\; (1+\alpha_t)\log p_{\mathcal{M}}(y) - \alpha_t \log p_{\varnothing}(y) \\ &=\; \log p_{\varnothing}(y) + (1+\alpha_t)\log \frac{p_{\mathcal{M}}(y)}{p_{\varnothing}(y)}, \end{aligned} \qquad (4)$$

and picks $y_t = \arg\max_{y \in \mathcal{V}_\beta} s_t(y)$ over the plausible set $\mathcal{V}_\beta = \{y : p_{\mathcal{M}}(y) \geq \beta \max_{y'} p_{\mathcal{M}}(y')\}$ with β=0.1, as in VCD [30], so that tokens the memory makes implausible are never promoted. Three choices adapt this rule to grounding.

No-video counterfactual. The negative memory is empty rather than noised or shuffled. It isolates exactly the prior measured in Sec. 3.3, and it is cheap: its cache holds only the prompt.

Time scope. PCD is applied only while a time value is written: inside a `Seen:` time before its description and inside `Time: [s, e]`. All other tokens, including the answer text, are decoded greedily from $p_{\mathcal{M}}$, so the answer is unaffected where the memory already speaks.

Confidence-adaptive strength. We set $\alpha_t = \alpha_{\max}\big(1 - \max_y p_{\mathcal{M}}(y)\big)$ with $\alpha_{\max}$=1. Where the memory makes the model confident (*e.g.* evidence that truly starts at 0 s), PCD fades and the early answer is kept; where the model hesitates between times, the evidence term decides.

The timeline prompt matters: PCD acts on the listed `Seen:` moments and then on the interval, so the interval is written from contrasted observations. When the answer is a single interval, PCD has little to correct: with the one-line grounding prompt it changes mIoU by -0.1 (n.s., Sec. 6.2), and with the direct-answer prompt by $+0.4$.

4.3. BOLT: question-guided frames

When the window is much longer than F frames, which frames are encoded can matter as much as which tokens are kept. As a front end we sample frames as in BOLT [34]. Each second of video is embedded once with SigLIP 2 [45], which gives a causal 1 fps index. The relevance z_j of second $j \leq q$ is its z-scored cosine similarity with the question, and the F frames are taken at evenly spaced quantiles of

$$p(j) \;=\; \tfrac{1}{2}\cdot\tfrac{1}{q} \;+\; \tfrac{1}{2}\,\mathrm{softmax}(z)_j, \qquad (5)$$

so that half of the frames stay uniform. BOLT is skipped when uniform frames would be less than 1 s apart. Once tokens are chosen by the question, BOLT adds only $+0.8$ to $+0.9$ mIoU (not significant on S-EMBER, QVHighlights or ActivityNet); the main-table runs include it.

5. Experimental Setup

Models. Qwen3-VL-8B-Instruct [38], 64 frames sampled uniformly from $[0, q]$ (or by BOLT, Sec. 4.3) at native resolution, greedy decoding or PCD, at most 384 new tokens, batch size 8. Unless stated otherwise all methods share the timeline prompt. In Sec. 6.2 we add TimeLens-8B [50], a Qwen3-VL-8B model fine-tuned for grounding, and the one-line prompt of TimeLens-Bench. Batch size changes about half of the greedy outputs and up to about two points of a score; comparisons with the uncompressed model, which runs at batch size 1–2, carry this noise.

Datasets. **S-EMBER** grounded QA [47]: 475 questions (191 duration, 184 counting, 100 location) on the first 300 videos, asked mid-stream; the memory contains only the frames before the question. **QVHighlights** [29] val: 500 queries with a single relevant window (a seed-0 sample of the 1,020 single-window queries), videos of about 150 s. **ActivityNet Captions** [28] val_1: 500 queries, one sentence per video, videos of about 115 s. **Charades-STA** [22] test: 500 queries, videos of about 30 s. For the last three, the question is *"When does the following happen in the video: <query>?"* and the window is the whole video. **TimeLens-Bench** [50]: the re-annotated Charades-STA, QVHighlights and ActivityNet sets, evaluated with their one-line prompt (full sets and a 513-query QVHighlights subset). **NExT-GQA** [49] test: 1,005 questions on about 40 s videos, answer and interval asked in two separate queries. Our QVHighlights and ActivityNet copies are low-resolution re-encodes (153 and about 84 visual tokens per frame pair, against 660 on S-EMBER), so we also report budgets in absolute tokens.

Budget. 10% of the visual tokens of the 64-frame memory, timestamps included (S-EMBER: about 2,140 tokens; QVHighlights: about 520). ActivityNet is reported at 30% (400–900 tokens), because at 10% the memory holds only 120–300 tokens (Sec. 6.6).

Baselines. All tokens (no pruning); random tokens; FastVID [41], VidCom^2 [35], FlashVID [18] (vision-side stages) and SemVID [14], re-implemented and checked to keep the same tokens as the authors' code on test inputs; each with and without PCD. **[TODO: SnapKV-style observation-window eviction, PruneVid, global question top-k without quotas, and FV-Action / attention readout on the same memory.]**

Floor-adjusted evaluation. For every dataset we report three video-free or video-light references. (i) The *best constant interval* $[aq, bq]$ over a 20×20 grid, tuned on the test questions, which upper-bounds any constant rule. (ii) *Length rescaling*: the no-PCD prediction stretched about its

start by a factor fitted once on S-EMBER (1.76) and applied unchanged to the other datasets; it tests whether a gain could come from longer intervals alone. (iii) The split into *closed* questions (gold end $< 0.8q$) and the rest. End-snapping on S-EMBER is discussed in Appendix D.

Metrics and statistics. mIoU and R@$\{0.3, 0.5, 0.7\}$; on S-EMBER also answer accuracy and GQ@0.5, judged with the official S-EMBER judge prompt by a local Qwen3.8-27B-FP8 model, all rows of a table in one judging session. Answers that state times as minutes:seconds are converted to seconds for every method alike (Appendix A). We use a paired bootstrap over questions (2,000 resamples) and mark with * differences whose 95% interval excludes 0. **[TODO: video-clustered bootstrap.]**

Protocol. All design choices (scores, quotas, α, scope) were made on S-EMBER. The configuration was then frozen and run once on QVHighlights, ActivityNet and Charades-STA. **[TODO: held-out S-EMBER videos as a test split.]**

6. Results

We organise the results by question. Section 6.1 compares BOCC with all baselines and floors on four datasets with the timeline answer format, and Sec. 6.2 repeats the key comparisons under the standard one-line prompt, on a grounding-tuned model and on NExT-GQA. Section 6.3 shows how published pruners behave on grounding. Sections 6.4 and 6.5 analyse the two components. Sections 6.6 and 6.7 study the budget and how to spend it. Section 6.8 compares with upper bounds that use the gold interval, Sec. 6.9 analyses BOLT, Sec. 6.10 lists what did not work, and Sec. 6.11 analyses the remaining errors.

Reading the tables. Numbers are mIoU, or mIoU / R@0.5 where two are given. Δ is a paired difference; * marks differences whose 95% paired bootstrap interval excludes 0. Tables built from the early S-EMBER phases (Tabs. 17 to 20 and 25 and Tab. 2) use an earlier implementation of PCD, the official parser and batch size 1; on S-EMBER the two parsers differ by at most 0.7 mIoU and batch size changes no paired result. In every table the best value is in **bold** and the second best underlined, per column unless a caption says per row; floors and differences (Δ) are not ranked.

6.1. Main results

Tables 4 and 5 lead to three conclusions.

(i) With the timeline prompt, BOCC nearly doubles the uncompressed model on the least biased benchmark. On QVHighlights, with 10% of the tokens, BOCC reaches 54.8 R@0.5 (R@0.7: 30.2) against 28.6 (18.4) for all tokens and 25.0 (16.2) for random tokens, and clears every floor by a wide margin (+22.1 mIoU over the best constant, +8.6 over length rescaling). This comparison holds the prompt fixed; the one-line prompt of TimeLens-Bench gives the uncompressed model far higher scores, and there the gain of BOCC is smaller (Sec. 6.2).

(ii) On S-EMBER, the gain is real where the window end is not a shortcut. BOCC beats all tokens, random selection and SemVID under the same decoding. Against the floors its pooled gain is small and not significant, because two thirds of S-EMBER intervals end at the question time. On closed intervals, where that shortcut does not exist, BOCC gains $+7.6^*$ mIoU over the best constant and $+7.5^*$ over length rescaling (Tab. 26).

(iii) On ActivityNet and Charades-STA, the timeline prompt is the bottleneck. With this prompt and our low-resolution copies, the uncompressed model is far below the constant floor on both (22.9 *vs*. 37.9 and 16.6 *vs*. 29.7); with the one-line prompt of TimeLens-Bench it scores 47.1 and 48.5 mIoU (Sec. 6.2). BOCC improves on all tokens and on every pruner on ActivityNet, but no method reaches the floor, and on Charades-STA Coverage is worse than random selection (Sec. 6.7). We report these as limits, not successes.

Answers. BOCC also improves the answers (Tab. 6): accuracy 21.7 *vs*. 17.3 for random tokens ($+4.4^*$) and 19.8 for all tokens, and GQ@0.5 13.3 *vs*. 7.4 and 8.8. Coverage alone gives the best accuracy (22.5); PCD trades about one point of accuracy for its grounding gain, and differences below about two points of accuracy are within the decoding noise of the judge setup. **[TODO: SemVID + PCD in the same judging session.]**

Question types. BOCC improves duration and counting questions most (Tab. 7). Location questions ("where did I put …") remain essentially unsolved by every method (mIoU $\leq$ 11): 62 of the 100 location questions have annotators who disagree on the interval (mean pairwise IoU < 0.6), which caps any method.

6.2. Standard prompt, a grounding-tuned model, and short videos

The results above use one model and the reasoning-style timeline prompt on every dataset. We now test the method under the standard one-line grounding prompt of TimeLens-Bench [50], on a grounding-tuned model (TimeLens-8B [50]), and on NExT-GQA [49]. Unless stated otherwise, paired differences use the same questions; unpruned runs use batch size 1–2 and pruned runs batch size 8, which alone moves scores by up to about two points (Sec. 5).

Our pipeline reproduces published numbers. With the TimeLens evaluation script, Qwen3-VL-8B scores 48.5, 59.6 and 47.1 mIoU on the full Charades-STA, QVHighlights and ActivityNet sets of TimeLens-Bench (published: 48.3, 59.4, 46.8), and TimeLens-8B 55.4, 65.7 and 53.2 (published: 55.2, 65.5, 53.2). On the same queries, our own pipeline without pruning matches the script: +0.5

Table 4. **Grounding with 10% of the visual tokens** (ActivityNet: 30%), mIoU / R@0.5. Grey rows are floors: the best video-free constant interval, tuned on the test questions, and the BOLT + Coverage prediction without PCD lengthened by a factor fitted on S-EMBER. Coverage rows include BOLT, which is skipped on Charades-STA, where uniform frames are already dense. Empty cells: not yet computed. S-EMBER closed intervals: Tab. 26.

Method	S-EMBER	QVHighlights	ActivityNet (30%)	Charades-STA
All tokens (no pruning)	28.0 / 26.7	29.1 / 28.6	22.9 / 18.2	16.6 / 14.2
Random tokens	25.6 / 24.6	26.9 / 25.0	22.9 / 18.6	17.4 / 13.8
FastVID [41]	17.9 / 15.8	19.2 / 15.4		13.3 / 11.6
VidCom^2 [35]	20.4 / 18.7	19.9 / 18.2		12.3 / 8.4
FlashVID [18]	22.7 / 21.5	22.8 / 20.4	20.5 / 15.4†	14.1 / 10.8
SemVID [14]	23.3 / 20.6	24.5 / 21.2	22.8 / 18.6	14.6 / 11.2
BOLT + Coverage	32.5 / 34.1	34.5 / 35.6	25.4 / 21.2	16.4 / 13.6
BOLT + Coverage + PCD	**39.3 / 40.8**	**48.8 / 54.8**	**34.7 / 32.4**	23.3 / 21.2

† at 10%.

Table 5. **Paired gains of BOCC in mIoU** (*: 95% paired bootstrap interval excludes 0).

BOCC *vs.*	S-EMBER	QVHighlights	ActivityNet (30%)	Charades-STA
All tokens	$+11.2^*$	$+19.7^*$	$+11.8^*$	
Random + PCD	$+4.2^*$	$+11.0^*$	$+3.3^*$	-2.9^*
SemVID + PCD	$+4.5^*$	$+14.1^*$	$+4.5^*$	
Best constant	$+2.4$	$+22.1^*$	-3.1^*	-6.4^*
Length rescaling	$+1.0$	$+8.6^*$	$+0.8$	-0.7

Table 6. **Grounded QA on S-EMBER** (10%; one judging session). ΔAcc. *vs.* random tokens.

Method	Acc.	GQ@0.5	mIoU	ΔAcc.
All tokens	19.8	8.8	28.0	+2.5
Random tokens	17.3	7.4	25.6	–
FastVID	15.2	4.8	17.9	−2.1
VidCom^2	16.2	5.9	20.4	−1.1
FlashVID	17.3	6.1	22.7	+0.0
SemVID	19.8	7.6	23.3	+2.5
Coverage, uniform frames	**22.5**	11.2	31.7	$+5.3^*$
Random + PCD	20.4	12.2	35.0	+3.2
SemVID + PCD$^\circ$	17.9	9.9	32.3	+0.6
BOCC	21.7	13.3	**39.3**	$+4.4^*$
BOCC + coverage gate	22.1	**13.5**	**39.3**	$+4.8^*$

$^\circ$ earlier implementation of PCD.

Table 7. **S-EMBER by question type**, mIoU / R@0.5 (10%).

Method	Duration (191)	Counting (184)	Location (100)
All tokens	37.6 / 40.3	30.1 / 25.5	5.9 / 3.0
Random tokens	33.2 / 35.1	29.6 / 26.1	3.5 / 2.0
BOLT + Coverage	43.2 / 46.6	35.3 / 36.4	7.1 / 6.0
Random + PCD	41.0 / 40.8	41.9 / 40.2	**11.0 / 9.0**
SemVID + PCD	38.9 / 36.6	44.6 / **50.0**	8.9 / 8.0
BOCC	**47.6 / 50.8**	**46.9** / 48.4	9.3 / 8.0

Table 8. **Answer format** (QVHighlights, 513 TimeLens-Bench queries, Qwen3-VL-8B, 300 frames, 31% of the tokens), mIoU. One-line: the TimeLens-Bench prompt. Timeline: the prompt of all other tables.

	One-line prompt		Timeline prompt	
Selection	greedy	PCD	greedy	PCD
All tokens	60.5	–	30.4	–
Random tokens	58.2	–	**33.4**	43.9
Coverage	**61.7**	61.6	30.7	43.9

(Charades, 400 queries), +1.4 (ActivityNet, 400) and 0.0 (QVHighlights, 513) mIoU, none significant (Tab. 27).

The timeline prompt is costly for single-moment grounding. Table 8 changes only the prompt. On QVHighlights queries, asking for a timeline of observations before the interval costs about 30 mIoU against the one-line prompt (-30.0^* without pruning, -31.0^* for Coverage). PCD recovers part of the loss ($+13.2^*$ on Coverage, $+10.4^*$ on random tokens) but stays far below the one-line prompt (-17.8^*), and under the one-line prompt it has no effect (-0.1, n.s.; also -0.3 and -0.5 with BOLT). With the timeline prompt, Coverage is not better than random selection on these queries (-2.8^* greedy, $+0.1$ with PCD). Two consequences follow. First, the QVHighlights, ActivityNet and Charades-STA numbers of Tab. 4 are relative to a format

Table 9. **Standard grounding prompt** (QVHighlights, 513 TimeLens-Bench queries, one-line prompt), mIoU / R@0.5. Top: 300 frames at protocol size; bottom: 64 frames chosen by BOLT. Kept: visual tokens in memory. TimeLens input: the TimeLens-Bench input re-sampled to the same number of tokens. Keep ratios count timestamp tokens too: 31% of all tokens is 24% of the visual tokens.

Method	Kept	Qwen3-VL-8B	TimeLens-8B
All tokens	13,650	60.5 / 64.9	67.1 / 73.1
TimeLens input, 2 fps low-res	4,200	58.0 / 62.0	64.3 / 68.8
Random tokens (31%)	3,271	58.2 / 60.8	65.3 / 70.4
SemVID (31%)	4,232	58.8 / 62.4	–
Coverage (31%)	3,271	**61.7 / 66.7**	**67.3 / 73.9**
TimeLens input, 1 fps low-res	1,125	49.5 / 53.0	57.5 / 61.8
BOLT + random (23%)	897	55.5 / 60.8	–
BOLT + Coverage (23%)	897	**58.3 / 63.7**	**64.3 / 71.2**

Table 10. **Full TimeLens-Bench sets** (one-line prompt, 31% of the tokens), mIoU / R@0.5. All tokens and low-res: the TimeLens evaluation script, the latter with a lower resolution at a matched budget (about 3.2k / 4.2k visual tokens on Charades-STA / QVHighlights). Paired ΔmIoU below.

	Qwen3-VL-8B		TimeLens-8B	
Method	Charades	QVHighlights	Charades	QVHighlights
All tokens (no pruning)	48.5 / **54.0**	59.6 / **64.5**	**55.4 / 63.1**	**65.7 / 71.8**
TimeLens input, low-res	46.0 / 49.6	57.3 / 61.6	52.8 / 59.6	63.3 / 68.4
Random tokens (31%)	48.2 / 52.3	58.1 / 62.6	54.5 / 61.7	64.2 / 69.6
Coverage (31%)	**48.6 / 54.0**	**59.7 / 64.5**	55.2 / **63.1**	65.4 / 71.0
Coverage *vs.* random	$+0.5^*$	$+1.6^*$	$+0.7^*$	$+1.2^*$
Coverage *vs.* low-res	$+2.6^*$	$+2.4^*$	$+2.4^*$	$+2.1^*$
Coverage *vs.* all tokens	+0.1	+0.0	−0.2	−0.3

that handicaps the uncompressed model; the gain over "all tokens" there (Sec. 6.1) does not hold against the one-line prompt. Second, PCD corrects time values in reasoning-style answers, where the model writes several times before the interval; when the answer is a single interval, there is little for it to correct.

Under the standard prompt, Coverage transfers to both models. With the one-line prompt (Tab. 9), Coverage at 24% of the visual tokens beats random selection on both models ($+3.5^*$ and $+2.0^*$ mIoU), beats SemVID ($+2.9^*$), and beats the TimeLens input re-sampled to a similar budget ($+3.7^*$, $+3.0^*$). It matches the uncompressed model ($+1.2$ for Qwen3-VL-8B, with a lower bound just at zero; $+0.2$ for TimeLens-8B). At 897 tokens, BOLT with Coverage exceeds a 1 fps TimeLens input of 1,125 tokens by $+8.8^*$ and $+6.8^*$, and BOLT with random tokens by $+2.8^*$. At this budget the grounding-tuned model does lose accuracy: TimeLens-8B with BOLT and Coverage is 2.8^* mIoU below its uncompressed memory, whereas at 3,271 tokens it is level ($+0.2$).

Table 11. **A grounding-tuned model on S-EMBER** (10%, timeline prompt), mIoU / R@0.5 and Acc. / GQ@0.5 (one judging session for all rows).

	Qwen3-VL-8B		TimeLens-8B	
Method	mIoU / R@0.5	Acc. / GQ	mIoU / R@0.5	Acc. / GQ
All tokens (no pruning)	28.0 / 26.7	19.6 / 8.8	33.1 / 34.1	20.6 / 12.2
Random tokens	25.6 / 24.6	17.1 / 7.2	28.2 / 26.7	19.6 / 9.5
Random + PCD	35.0 / 33.9	20.0 / 11.8	38.4 / 39.4	20.4 / 12.4
BOLT + Coverage	32.5 / 34.1	**22.3** / 11.2	37.2 / 37.9	22.9 / 13.1
BOCC	**39.3 / 40.8**	21.9 / **13.5**	**43.9 / 46.9**	**27.4 / 17.5**
BOCC *vs.* all tokens	$+11.2^*$	$+2.3$ / $+4.6^*$	$+10.7^*$	$+6.7^*$ / $+5.3^*$

The same holds on the full sets. On the complete Charades-STA (3,363 queries) and QVHighlights (1,541) sets of TimeLens-Bench (Tab. 10), Coverage at 31% of the tokens is lossless on both models (-0.3 to $+0.1$ mIoU, n.s.). It beats random selection in all four cells ($+0.5^*$ to $+1.6^*$) and the TimeLens input reduced to the same budget by lowering the resolution in all four ($+2.1^*$ to $+2.6^*$). The gap to random is largest on QVHighlights, where videos are longer; on Charades-STA, 31% leaves about 3,350 of 10,800 tokens and random selection is already close, as on NExT-GQA below. Keeping full resolution and choosing tokens beats spending the same budget on a lower resolution: random selection also beats the low-res input on Charades-STA ($+2.2^*$ and $+1.7^*$), though not significantly on QVHighlights.

On grounded QA, the full method transfers to a grounding-tuned model. S-EMBER answers explain before they ground, so the timeline format is natural there. TimeLens-8B, trained for grounding, is above Qwen3-VL-8B in every row of Tab. 11, and the ordering of the methods is the same for both models. With 10% of the tokens, BOCC improves on the uncompressed TimeLens-8B on every metric: $+10.7^*$ mIoU, $+6.7^*$ accuracy and $+5.3^*$ GQ@0.5, and on random selection by $+15.7^*$ mIoU, $+7.8^*$ accuracy and $+8.0^*$ GQ@0.5. Each component helps on its own: PCD adds $+10.2^*$ mIoU to random tokens and $+6.7^*$ to BOLT + Coverage, and BOLT + Coverage adds $+9.0^*$ to random tokens. With TimeLens-8B, BOCC answers and grounds better than Qwen3-VL-8B with the gold window of Tab. 23 (GQ@0.5 17.5 *vs.* 15.8).

On short videos, the gain of Coverage grows as the budget shrinks. NExT-GQA videos last about 40 s, so 64 frames already sample them densely (Tab. 12). The advantage of Coverage over random selection grows as the budget shrinks: on Qwen3-VL-8B it adds $+1.2$ (n.s.), $+3.1^*$ and $+6.8^*$ Acc@QA at 30%, 10% and 5%, and at 5% also $+5.2^*$ Acc@GQA. On TimeLens-8B the gains are $+3.1^*$ Acc@QA at 10%, and at 5% Coverage beats random selection on every metric ($+5.6^*$ Acc@QA, $+4.7^*$ Acc@GQA, $+2.4^*$ mIoP, $+1.6^*$ mIoU). At 30% Coverage is lossless ($+0.5$ Acc@GQA, -0.2 mIoU, n.s.) and grounds better than SemVID ($+1.8^*$ mIoU). At 5%, SemVID also answers

Table 12. **NExT-GQA, multiple choice** (test, 1,005 questions, 64 frames, two-query protocol: one query for the answer, one for the interval; no PCD). 5% keeps 406 of 8,122 tokens.

Model	Method	Acc@QA	Acc@GQA	mIoP	mIoU
Qwen3-VL-8B	All tokens	79.7	38.9	**48.0**	**37.7**
	Random 30%	78.6	38.6	47.5	37.3
	SemVID 30%	78.9	38.5	46.2	35.8
	Coverage 30%	**79.8**	**39.4**	47.9	37.6
	Random 10%	75.3	34.9	45.2	35.3
	Coverage 10%	78.4	36.1	44.6	34.8
	Random 5%	68.3	27.3	40.5	30.5
	SemVID 5%	74.0	30.0	41.0	30.0
	Coverage 5%	75.0	32.4	41.6	31.4
TimeLens-8B	All tokens	**76.7**	**41.5**	**52.1**	**39.8**
	Random 10%	72.3	36.9	48.9	37.7
	Coverage 10%	75.4	38.5	49.3	37.8
	Random 5%	66.0	28.7	43.6	34.0
	Coverage 5%	71.5	33.3	46.0	35.6

Table 13. **NExT-GQA, open-ended** (Qwen3-VL-8B, 1,005 questions, options removed). Answers are judged correct when they match the ground truth in meaning (LLM judge, Video-ChatGPT prompt [36]); Acc@GQA also requires IoP ≥ 0.5. Two-query: the answer prompt plus the TimeLens grounding query. Joint: one timeline answer with the interval.

Protocol	Method	Acc@QA	Acc@GQA	mIoP	mIoU
Two-query	All tokens	33.5	**17.4**	**48.0**	**37.7**
	Coverage 10%	33.0	16.4	44.6	34.8
Joint	All tokens	**34.1**	11.0	30.4	20.8
	Random 10%	31.1	9.7	30.5	19.9
	SemVID 10%	28.1	7.9	28.6	17.2
	Coverage 10%	33.2	10.6	30.6	20.4
	Coverage 10% + PCD	32.2	10.7	33.4	24.7

well (74.0, $+5.8^*$ over random), and Coverage leads it by $+2.5$ Acc@GQA (interval just including 0). When frames are dense, where tokens are kept matters most for the answer and less for the interval, as on Charades-STA (Sec. 6.7). We use the two-query protocol; asking for answer and interval in one reply, as in prior zero-shot reports [16], lowers Acc@GQA by 16.5^* (Qwen3-VL-8B) and 9.2^* (TimeLens-8B), mostly because the time values are then written as minutes:seconds or copied from the prompt's example, so absolute numbers are comparable only within a protocol.

Open-ended answers. Without the options (Tab. 13), answering is far harder (33.5 *vs*. 79.7 Acc@QA) while grounding is unchanged, since the grounding query is the same. Our judge accepts the ground-truth answer for all 1,005 questions and agrees with letter correctness on 99.2% of the multiple-choice answers. Without pruning, Qwen3-VL-8B reaches 17.4 Acc@GQA zero-shot, against 21.4 for TOGA [23], which is trained for open-ended grounded QA and judged with GPT-3.5. Coverage at 10% keeps the answers in both protocols (-0.5 and -0.9 Acc@QA, n.s.), whereas random selection and SemVID lose them (-3.0^*, -6.1^* in the joint protocol; Coverage leads SemVID by $+5.2^*$ Acc@QA and $+2.8^*$ Acc@GQA). The joint timeline answer grounds far worse than the separate query (20.8 *vs*. 37.7 mIoU): its listed moments start where the subject is first visible, so 81% of its intervals start at 0 s (ground truth: 19%). PCD raises its mIoU ($+3.9^*$) but not Acc@GQA, and on the two-query protocol PCD changes neither (-0.4 Acc@GQA, n.s.), as with the one-line prompt on QVHighlights. On NExT-GQA, the contribution is therefore Coverage; PCD helps only where the answer format lets the prior choose the times. **[TODO: two-query Random and SemVID 10% (being judged); TimeLens-8B open-ended.]**

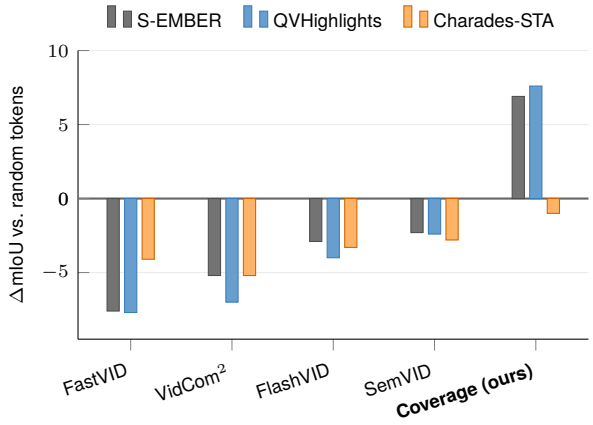


Figure 4. **QA pruners fall below random selection on grounding.** Paired mIoU difference to random token selection at the same 10% budget, greedy decoding. All four published pruners lose on all three datasets (all significant except SemVID on QVHighlights). Coverage gains on the two long-video datasets and is level with random selection on 30 s Charades videos.

6.3. QA pruners on grounding

Every published pruner (Fig. 4) falls below random token selection on S-EMBER (-2.3 to -7.6 mIoU, all significant), QVHighlights (-2.4 to -7.7; SemVID not significant) and Charades-STA (-2.8 to -5.2, all significant); on ActivityNet at 10% the input-level pruners are above random, which we discuss in Sec. 6.7. Random selection varies little across seeds (26.0 ± 0.6 mIoU over three seeds on S-EMBER), so the gaps are not noise. The ordering follows coverage (Tab. 1): the methods that empty frames most aggressively lose most, and those that spread tokens lose least. Accuracy, in contrast, is barely affected (Tab. 6): FastVID's accuracy is within noise of random selection (-2.1, n.s.) while its mIoU is 7.6 lower. A pruner built to keep answers keeps answers, not time. PCD helps every pruner ($+7$ to $+12$ mIoU) but does not close the gap to BOCC.

Table 14. **Component ladder** (mIoU / R@0.5) with 10% of tokens; PCD: prior-contrastive decoding.

Configuration	S-EMBER	QVHighlights	ActivityNet
Uniform, random tokens	25.6 / 24.6	26.9 / 25.0	22.9 / 18.6
BOLT, random tokens	29.4 / 29.5	28.7 / 28.0	–
Coverage (uniform frames)	31.7 / 32.4	33.7 / 33.0	24.6 / 21.6
BOLT + Coverage	32.5 / 34.1	34.5 / 35.6	25.4 / 21.2
BOLT + Coverage + PCD	**39.3 / 40.8**	**48.8 / 54.8**	**34.7 / 32.4**

Table 15. **Design choices of Coverage** (S-EMBER, greedy decoding). Frames: U = uniform. Δ *vs.* the equal quota with question top-k on the same frames.

Token selection	Frames	mIoU	Δ
Equal quota, question top-k	BOLT	32.5	–
Quota ∝ SigLIP relevance (floor 50%)	BOLT	**33.0**	+0.5
Quota ∝ SigLIP (floor 25%)	BOLT	32.6	+0.1
Quota ∝ SigLIP + attention	BOLT	32.9	+0.4
Quota ∝ attention mass	BOLT	32.1	−0.4
One token set shared by all layers	BOLT	31.9	−0.6
Equal quota, question top-k	U	31.7	–
Quota ∝ SigLIP relevance	U	31.4	−0.2
Equal quota, sampling instead of top-k	U	30.0	−1.7

6.4. Analysis of Coverage

Contribution. Table 14 adds the components one at a time. Coverage is the largest gain on the memory side: $+6.1^*$ mIoU on S-EMBER and $+6.8^*$ on QVHighlights over random tokens. BOLT adds $+0.8$ to $+0.9$ on top (not significant on any dataset); once tokens are chosen by the question, which frames are encoded matters less. **[TODO: Coverage + PCD without BOLT on all datasets.]**

Design choices. Table 15 varies one choice at a time. *The quota should be equal*: five relevance-weighted variants, from SigLIP similarity, attention mass or both, with floors of 25–50%, are within ± 0.5 mIoU of the equal quota and none is significant (the uniform-frame SigLIP variant was our pre-registered primary test). Quotas that follow raw attention mass are pulled by attention sinks and recency bias: a single frame received up to 227 of about 2,140 kept tokens. *The score should be the actual question*: HERMES's generic probe questions give 23.7 even with an equal quota, and 18.3 without. *Deterministic top-k beats sampling* from a coverage-floored density (31.7 *vs.* 30.0); with the quota in place, coverage is already guaranteed. A contrastive token score that subtracts the attention of a generic sentence, meant to cancel sinks, does not help.

Per-layer selection. Table 16 compares one token set per layer with a single set shared by all layers. Sharing does not change random selection, which rules out per-layer diversity as the reason random selection beats input-level pruners. It costs BOCC -1.5^* mIoU: different layers attend to different tokens, and question-driven selection benefits from letting each choose.

Table 16. **Per-layer *vs.* shared token sets** (S-EMBER, 10%). Δ = shared − per layer.

Method	Per layer	Shared	ΔmIoU
Random tokens	25.6	26.3	+0.7
Random + PCD	35.0	35.8	+0.7
BOLT + Coverage	32.5	31.9	−0.6
BOCC	**39.3**	**37.8**	−1.5

6.5. Analysis of PCD

In reasoning-style answers, PCD is a plug-in for any memory. Table 17 applies PCD to eight memories: three random budgets, the uncompressed model, HERMES and stratified pruning, and two HERMES streaming memories. It improves mIoU significantly on all eight ($+4.6$ to $+8.0$) and GQ@0.5 on six, roughly halves the early starts everywhere, and its gains are largest where the memory is smallest. With 10% of the tokens and random selection it already beats the uncompressed model (32.6 *vs.* 28.0). PCD also transfers to FastVID ($+7.6^*$ mIoU, leak 38% → 12%), the pruner that loses coverage most.

The no-video counterfactual is the right one. Table 18 changes only what the memory is contrasted against. The no-video input isolates the prior and removes it (early starts 38% → 19%). Noised frames, as in VCD, still show part of the video *and* still produce the prior, so the two runs disagree mainly about content and little of the prior is removed ($+0.5$ mIoU). Permuted timestamps leave the prior in place as well. A one-sided rule that only penalises tokens the video argues against is worse than PCD (-5.6^* and -4.8^*): without video the model is about 99% sure of "0", so even a well-supported "0" looks like evidence against. Boosting visual attention during decoding, or stating the bias in the instruction, leaves the prior in place.

Time scope and adaptive strength. Contrasting every token also changes words that have nothing to do with time: the answer text changed in 85% of questions, lists of more than eight moments rose from 12 to 56, and over-counting rose from 14% to 34% of counting questions. Restricting PCD to time values keeps the grounding gain and disturbs the text less (Tab. 19, top). PCD also risks hurting questions whose evidence really starts at 0 s. With a constant strength, their accuracy on the uncompressed memory falls from 21.8 to 16.7; with the adaptive strength, which fades when the model is confident, it rises to 23.1 (Tab. 19, bottom; $+6.4^*$ *vs.* constant). The adaptive rule also gives the best *discrimination*, early starts on early-evidence questions minus early starts on the others (56.4 *vs.* 49.6 for greedy,

Table 17. **PCD on eight memories** (S-EMBER, Acc. / mIoU / GQ@0.5; earlier implementation of PCD). Plain PCD: on all tokens with constant α=0.5; PCD: confidence-adaptive strength on time values only (ours). Δ PCD *vs.* greedy. Best and second best per row.

Memory	Greedy	Plain PCD	PCD	ΔmIoU	ΔGQ@0.5	Early starts
Random 5%	15.8 / 23.6 / 5.9	15.4 / 29.2 / 9.1	**16.4 / 31.5 / 9.9**	+8.0*	+4.0*	39% → 20%
Random 10%	15.2 / 25.8 / 7.2	**16.2** / 32.1 / **10.1**	**16.2 / 32.6 / 10.1**	+6.8*	+2.9	38% → 19%
Random 25%	17.7 / 28.2 / 8.2	**18.9** / 33.0 / 11.8	**18.9 / 34.9 / 13.3**	+6.7*	+5.1*	37% → 21%
All tokens	18.5 / 28.0 / 8.6	19.8 / 31.5 / 11.2	**21.5 / 32.7 / 13.3**	+4.6*	+4.6*	41% → 23%
HERMES 10%	13.1 / 18.3 / 4.0	**15.4** / 22.6 / **7.2**	13.3 / **24.1** / 5.9	+5.7*	+1.9	56% → 31%
Stratified 10%	14.1 / 23.7 / 6.1	14.5 / 29.9 / 8.8	**15.8 / 31.0 / 10.5**	+7.3*	+4.4*	36% → 17%
HERMES streaming 4k	13.5 / 21.6 / 4.0	**17.1** / 27.4 / **8.2**	15.4 / **28.6** / **8.2**	+6.9*	+4.2*	45% → 24%
HERMES streaming 6k	14.3 / 22.6 / 4.8	14.7 / 28.1 / 7.6	**15.4 / 29.8 / 8.4**	+7.2*	+3.6*	46% → 24%
Mean of 8	15.3 / 24.0 / 6.1	16.5 / 29.2 / 9.3	**16.6 / 30.7 / 10.0**	+6.7	+3.9	

Table 18. **What to contrast against** (S-EMBER, random 10%).

Counterfactual	mIoU	GQ@0.5	Early
None (greedy)	25.8	7.2	38%
No video (ours)	**32.1** (+6.2*)	**10.1** (+2.9*)	**19%**
Noised frames (VCD [30])	26.3 (+0.5)	7.8 (+0.6)	30%
Evidence-against rule, α=1 / 2	26.5 / 27.3	–	–
Attention gain (PAI-style)	+1.3 (n.s.)	+0.8	42%
Instruction "do not assume early"	25.7	6.9	37%

Table 19. **Time scope and adaptive strength** (S-EMBER, Acc. / mIoU / GQ@0.5). Top: PCD on all tokens *vs.* time values only. Bottom: the 78 questions whose evidence truly starts in the first 5% of the window. Top: best per row.

Memory	All tokens	Time values only
Random 10%	**16.2** / 32.1 / 10.1	16.0 / 32.1 / **10.3**
Random 25%	**18.9** / 33.0 / 11.8	17.7 / **34.0** / 11.8
All tokens	19.8 / 31.5 / 11.2	**20.8 / 32.9 / 12.4**
HERMES 10%	**15.4** / 22.6 / 7.2	14.7 / **24.3** / **7.4**
Streaming 4k	**17.1** / 27.4 / 8.2	16.2 / **29.2** / **9.3**
Early-evidence questions (78), all tokens memory		
Greedy	21.8 / 49.2 / 16.7	
PCD, constant α	16.7 / 55.3 / 12.8	
PCD, adaptive α	**23.1 / 56.9 / 19.2**	

Table 20. **Decision rules replayed at the first time value** (S-EMBER, random 10%). Share of answers starting at "0".

Rule	Early evidence ↑	Other (leak) ↓
Greedy	**69%**	16%
PCD, α=0.5	46%	**6%**
PCD, adaptive	51%	7%
Evidence-against, α=4	51%	9%
Skewed or tempered prior	50–55%	8–11%
Best threshold (uses gold)	46%	**6%**

all-tokens memory).

PCD is near the limit of decoding. We logged both distributions at the first time value of every answer and replayed other decision rules offline (Tab. 20). Even a threshold tuned on the gold labels cannot beat PCD's trade-off between keeping true early starts and removing leaked ones. The remaining early-start errors are cases where the memory itself is unsure; they need evidence, not a better decoding rule.

Is PCD more than a longer interval? PCD lengthens predicted intervals (median predicted/gold length 0.57 → 0.67 on S-EMBER), and longer intervals alone raise mIoU on benchmarks with long gold intervals. The length-rescaling floor tests this directly (Tab. 5, last row). On QVHighlights (+8.6*) and on S-EMBER closed intervals (+7.5*, Tab. 26), BOCC clearly beats rescaling: PCD moves the interval toward the evidence, not just outward. On pooled S-EMBER, ActivityNet and Charades-STA the two are statistically tied, so on those sets the benefit of PCD cannot be separated from a calibrated lengthening. **[TODO: start/end error decomposition on closed questions; mismatched-video and shuffled-frame negatives.]**

Prompt format. PCD needs a reasoning-style answer: with the official direct-answer prompt it does not help (+0.4 mIoU). An *event* timeline that lists start and end per occurrence (`Event: 12.0 -- 15.5 seconds`) improves the greedy baseline by +7.4* mIoU by stating extents, and PCD still adds +2.1 to +3.3 on top (35.0 mIoU, 12.6 GQ@0.5 at 768 tokens). The two fixes overlap: the format corrects extents, PCD corrects the leak. BOCC uses the simpler timeline in all main results. **[TODO: event timeline with BOCC.]**

When does PCD help? The results so far point to one rule: PCD helps in proportion to how much the video-free prior decides the time that is written. Its gain is largest where the memory says least (random 5%: +8.0* mIoU; uncompressed: +4.6*, Tab. 17); the adaptive strength, which fades

when the memory makes the model confident, protects true early starts that a constant strength removes (Tab. 19); and with a one-line answer, where the model states a single interval instead of listing observations, PCD has no effect at all (Sec. 6.2). We state this as a hypothesis that the per-token log-probabilities we already store can test directly. **[TODO: PCD gain against the divergence between $p_{\mathcal{M}}$ and $p_{\varnothing}$ at each written time value, across all conditions; one prediction stated before the run.]**

6.6. Budget

Table 21 and Fig. 5 vary the budget at 64 frames and show two regimes. Above roughly 500 kept tokens, question-driven selection is flat or even improves as the budget *shrinks*: on S-EMBER, BOLT + Coverage rises from 31.0 (20%) to 34.6 (5%) and BOCC from 36.8 to 39.3, while random selection and SemVID degrade. Pruning by the question removes distractors as well as cost: at 5% BOCC leads SemVID + PCD by $+7.4^{*}$ mIoU, at 20% by $+1.5$ (n.s.). Below roughly 300 kept tokens (QVHighlights 5%, ActivityNet 10%), the equal quota leaves only 2–4 tokens per frame pair, the top-attended tokens carry little content, and BOCC collapses (9.7 and 11.6), below random selection and below SemVID, whose cross-frame merging degrades more gracefully. The threshold is absolute, not relative: it depends on the number of tokens per frame, which is low in our re-encoded QVHighlights and ActivityNet copies.

6.7. Fewer frames or fewer tokens per frame?

The collapse at small budgets is not a limit of the equal quota itself but of how the budget is spent. In Tab. 22 every row of a block keeps the same number of tokens and only the split between frames and tokens per frame changes. With 32 frames instead of 64, each frame pair keeps twice as many tokens, and BOCC recovers from 9.7 to 43.1 mIoU on QVHighlights and from 11.6 to 33.9 on ActivityNet, where it now beats SemVID + PCD ($+5.5$) and nearly matches its own 64-frame result at three times the budget (34.7). Halving again to 16 frames loses a little: past a point, temporal coverage becomes the bottleneck. The baselines gain from fewer frames too, but less. On Charades-STA (30 s videos) fewer frames do not help any method. We therefore recommend the rule of Sec. 4.1: keep at least about eight tokens per frame pair and reduce F when the budget does not allow it. At equal tokens, more frames with fewer tokens each does not help either: 128 frames at 5% on S-EMBER give $+1.2$ mIoU (n.s.) without PCD and -3.6^{*} with it, with 74–78 unparseable answers.

6.8. Comparison with gold-informed upper bounds

Table 23 gives the memory information it should not have. At the same budget, BOCC without any gold information beats perfect token selection: over the oracle selector with PCD it gains $+5.1^{*}$ mIoU and $+4.2^{*}$ GQ@0.5. Knowing which tokens to keep does not stop the prior, and keeping only the gold frames' tokens starves the rest of the timeline. Against the strongest oracle, which sees only the evidence and also uses PCD, BOCC is level on grounding (-2.8 mIoU, -1.3 GQ@0.5, both n.s.); the remaining gap is in the answers (-5.3 Acc., interval just reaching 0). Even the gold window leaks 45%: with every frame inside the evidence, the model still often starts near 0 s. PCD lowers this to 35% and adds $+5.5^{*}$ mIoU on the gold window, but does not change its answers (-2.3, n.s.): more evidence raises accuracy, while PCD fixes the time values. The two levers are complementary.

6.9. Analysis of BOLT

The SigLIP 2 index does find the evidence: over the seconds before each question, the AUROC of evidence seconds against the rest is 0.663, and the top-ranked second lies in the gold interval for 63% of questions (chance 36%) and 41% of location questions (chance 13.5%). BOLT frames put 40% of the memory inside the evidence, against 36% for uniform frames. This helps random tokens ($+3.8^{*}$ mIoU) but adds only $+0.8$ to $+0.9$ (n.s.) once tokens are chosen by the question, because both stages focus the memory on the question. Subtracting a question-free prior from SigLIP's score does not help (AUROC 0.635–0.666), as expected for a retriever trained with a contrastive loss, whose score already estimates pointwise mutual information [45]. Re-weighting the token budget of uniform frames by the same relevance adds $+1.3$ (n.s.): which frames are encoded matters more than how the budget is split among fixed frames.

6.10. What did not work

Table 24 collects the ideas that gave little or nothing. They share a lesson: at a fixed budget, redistributing evidence (zooming, re-weighting frames, unequal quotas) is close to its ceiling, and only removing the prior where it acts (PCD) or guaranteeing coverage (the equal quota) gives large gains.

6.11. Where answers still fail

Table 25 shows that PCD works where it was aimed: intervals entirely before the gold one, the signature of the prior, fall from the largest error type to less than a fifth of answers. The remaining failures point to four limits. (i) *Answers, not intervals*: twice as many answers are well grounded but wrong as grounded and right; counting answers do not use the events the model itself listed, and an IoU of 0.5 still allows a two-fold duration error. (ii) *Recent events*: 81% of the intervals that still end before the gold one have evidence in the second half of the window, where an earlier similar event is picked; GQ@0.5 is 4.4 when the evidence lies in the last third of the window. (iii) *Sparse evidence*: with fewer than 4 of the 64 frames inside the evidence, GQ@0.5 is 1–2% for

Table 21. **Budget sweep at 64 frames** (mIoU). The second row gives the number of kept tokens per memory.

	S-EMBER			QVHighlights			ActivityNet		
Budget	5%	10%	20%	5%	10%	20%	10%	30%	50%
Kept tokens	∼1,070	∼2,140	∼4,280	∼260	∼520	∼1,040	120–300	400–900	600–1,500
Random	23.9	25.6	27.4		26.9		17.8	22.9	23.5
SemVID	20.9	23.3	26.5		24.5		21.2	22.8	22.3
R + C (no PCD)	34.6	32.5	31.0		34.5		11.3	25.4	25.0
SemVID + PCD	31.9	34.8	35.3		34.8		**26.8**	30.2	31.4
Random + PCD	34.5	35.0	36.5		37.8		21.5	31.4	32.2
BOCC	**39.3**	**39.3**	**36.8**	9.7	**48.8**	49.4	11.6	**34.7**	**34.4**

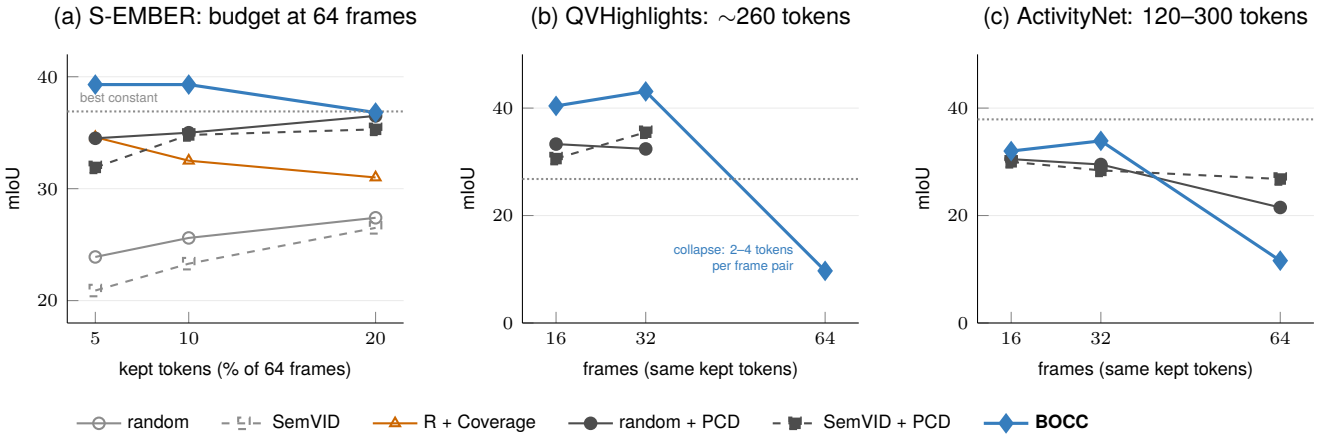


Figure 5. **Budget and how to spend it. (a)** At 64 frames, question-driven selection improves as the budget shrinks while random selection and SemVID degrade; BOCC leads at every budget that leaves at least about eight tokens per frame pair, and collapses below it (QVHighlights 5%, ActivityNet 10%). **(b, c)** At a fixed, very small number of kept tokens, 64 frames leave 2–4 tokens per frame pair and BOCC collapses; spending the same tokens on 32 frames recovers it (QVHighlights 9.7 → 43.1, ActivityNet 11.6 → 33.9). Dotted grey: best video-free constant.

any method, against 15% with 16 or more; no decoding rule recovers evidence that is not in the input. (iv) *Label ambiguity*: on the 98 questions whose annotators disagree (mean pairwise IoU < 0.6), every method scores near zero.

7. Limitations

Answer format. The main tables use a reasoning-style timeline prompt. On single-moment grounding it costs the uncompressed model about 30 mIoU against the standard one-line prompt, and with it Qwen3-VL-8B stays below the best constant interval on ActivityNet Captions and Charades-STA. Our gains over the uncompressed model on QVHighlights, ActivityNet and Charades-STA (Tab. 4) are therefore relative to this format; under the one-line prompt the comparison is Tab. 9 and, on the full Charades-STA and QVHighlights sets, Tab. 10; the full ActivityNet set is not yet run. On S-EMBER, BOCC beats the uncompressed model when both answer in the timeline format; the uncompressed model under the official direct-answer prompt is still to be reported. **[TODO: S-EMBER without pruning under the direct-answer and one-line prompts.]**

Where the prior is measured. Every measurement of the video-free prior (Sec. 3.3) uses the timeline format. PCD has no effect with a one-line answer, so the prior may decide few of the times written in that format, or none; we have not yet measured the no-video answers in it.

Minimum tokens per frame. Equal quotas fail when a frame pair keeps only 2–4 tokens. Spending the budget on fewer frames avoids this (Sec. 6.7), but the rule we use (at least about eight tokens per frame pair) was derived from the same results and still needs held-out validation.

Short, densely sampled videos. When uniform frames are less than a second apart (Charades-STA) or 64 frames already cover a 40 s video (NExT-GQA), question-driven selection gains little over random selection at 10–30% of the tokens; its advantage appears at tighter budgets (5% on NExT-GQA). On these videos PCD improves the interval only in the timeline format, and there it does not raise

Table 22. **Spending a fixed token budget on fewer frames** (mIoU / R@0.5, all with PCD). Within each block every row keeps the same number of tokens: F frames at keep ratio $\rho \cdot 64/F$. Best and second best per row.

Dataset (kept tokens)	Frames × keep ratio	Random + PCD	SemVID + PCD	BOCC
QVHighlights (∼260)	64 × 5%	–	–	9.7 / 6.6
	32 × 10%	32.4 / 29.2	35.5 / 33.6	**43.1 / 46.6**
	16 × 20%	33.3 / 32.4	30.6 / 28.0	**40.4 / 41.0**
ActivityNet (120–300)	64 × 10%	21.5 / 18.2	**26.8 / 23.6**	11.6 / 7.6
	32 × 20%	29.5 / 26.8	28.4 / 25.8	**33.9 / 32.0**
	16 × 40%	30.5 / 26.8	30.0 / 27.6	**32.0 / 29.8**
Charades-STA (∼10% of 64)	64 × 10%	**26.2 / 24.0**	21.7 / 19.2	23.3 / 21.2
	32 × 20%	**24.6 / 22.0**	20.8 / 16.0	23.6 / 21.0[§]

[§] Coverage on uniform frames; BOLT is skipped on Charades-STA.

Table 23. **Upper bounds that use the gold interval** (S-EMBER, Qwen3-VL-8B, 475 questions, one judging session). Oracle selector: the gold frames' tokens are kept first. Gold window: all 64 frames inside the gold interval. Leak: intervals starting in the first 5% of the window, on the 397 questions whose evidence starts later.

Memory	Gold	Acc.	mIoU	R@0.5	GQ@0.5	Leak
Random 10%	–	17.1	25.6	24.6	7.2	29%
Random 10% + PCD	–	20.0	34.4	33.3	11.8	14%
Oracle selector 10%	✓	18.9	27.9	25.7	9.3	33%
Oracle selector 10% + PCD	✓	18.5	33.5	33.7	9.3	16%
BOCC 10%	–	21.9	38.6	40.2	13.5	**11%**
All tokens	–	19.6	28.0	26.7	8.8	33%
Gold window	✓	**29.5**	35.9	37.9	**15.8**	45%
Gold window + PCD	✓	27.2	**41.4**	**42.9**	14.7	35%

Table 24. **Negative results** (S-EMBER; ΔmIoU against the stated reference).

Idea	Reference	ΔmIoU
VCD-style noised-frame contrast	greedy	+0.5
Permuted-timestamp contrast	greedy	+0.9
Evidence-against decoding rule	PCD	−5.6*
Attention gain while decoding	greedy	+1.3
Bias stated in the instruction	greedy	−0.1
Relevance-guided frame zoom	greedy	−1.5
Zoom inside the gold window	greedy	+2.3
Unequal per-frame quotas (5 variants)	equal quota	−0.4 to +0.5
Contrastive token score	question top-k	−1.2 to +0.5
Prior-subtracted retrieval	SigLIP	AUROC ≤ +0.003
128 frames at the same budget	64 frames	+1.2 / −3.6*

grounded accuracy (Tab. 13).

Not a streaming memory. The memory is rebuilt per question from the frames before the question. It is causal, but the full memory is prefilled once per question, so peak memory and time-to-first-token are those of the uncompressed model: BOCC reduces the memory the decoder reads, not the cost of building it. PCD also decodes a second, prompt-only row at every step. **[TODO: cost table; amortisation over several questions per video; write-once memory re-ranked at question time.]**

Table 25. **Error types** on S-EMBER, pooled over eight memories, greedy *vs*. PCD (earlier implementation of PCD).

Outcome	Greedy	PCD	Δ
IoU ≥ 0.5, answer correct	6.1%	9.9%	+3.8*
IoU ≥ 0.5, answer wrong	16.3%	20.7%	+4.4*
Partial overlap (IoU < 0.5)	17.0%	15.7%	−1.3
Inside gold, too short	22.2%	18.5%	−3.7*
Entirely *before* gold	33.1%	18.9%	−14.2*
Entirely after gold	2.7%	5.7%	+3.0*
Covers gold, too long	2.2%	2.5%	+0.3
No answer / unparsed	0.5%	8.1%	+7.6*

Scope of the evaluation. Our two models come from one family: TimeLens-8B is fine-tuned from Qwen3-VL-8B, and both read time from text timestamps. Most diagnostic results use 475–500-question subsets, and answers are judged by a local model with the official prompt. Every hyperparameter was chosen on the S-EMBER questions; the other datasets were run once with the frozen configuration, but a held-out S-EMBER split is still to be evaluated. **[TODO: a second model family; held-out S-EMBER split.]**

8. Conclusion

Pruning for QA keeps what is salient; grounding needs what is everywhere. A video LLM reads time from anchors that survive only with their frames, so pruners built for QA, which empty unremarkable frames, fall below random selection on grounding. Keeping one anchor in every frame, filled with what the question attends to, is enough to match the uncompressed model with a quarter of the visual tokens under the standard grounding prompt, on a general and a grounding-tuned model. When the answer lists what was

seen before it states the interval, as grounded QA does, a second failure appears: where the memory is silent, a video-free prior writes the times. Contrasting each time value with the model's own no-video prediction removes it, and with a tenth of the tokens BOCC then grounds better than the uncompressed model answering in the same format. The answer format itself turns out to be a first-order variable, worth about 30 mIoU on single-moment grounding, and with video-free floors it belongs in every evaluation of compressed video memories.

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A. Parsing times written as minutes:seconds

On videos longer than a minute, the model sometimes writes times as minutes:seconds, *e.g.* `Time: [1:45, 2:07]`, which the official S-EMBER parser scores as missing. We convert `h:mm:ss` and `m:ss` to seconds for every method alike. Method rankings are unchanged with the official parser. The main numbers with the official parser are: QVHighlights, BOCC 44.6 / 50.2 (*vs.* 48.8 / 54.8); ActivityNet at 30%, 33.3 / 31.0 (*vs.* 34.7 / 32.4). **[TODO: full table with the official parser.]**

B. S-EMBER closed intervals

Two thirds of S-EMBER gold intervals end at, or close to, the question time, so a constant interval ending at q is a strong floor on the pooled set. Table 26 restricts the main comparison to the *closed* questions (gold end $< 0.8q$, 33% of the questions), where that shortcut does not exist; the floors are re-tuned on this subset.

Table 26. **S-EMBER closed intervals** (10%, mIoU / R@0.5; settings as in Tab. 4). Right: paired ΔmIoU of BOCC (*: 95% paired bootstrap interval excludes 0).

Method	mIoU / R@0.5	BOCC *vs.*
Best constant interval	*32.1 / 29.9*	+7.6*
Length rescaling of $R + C$	*32.3 / 33.1*	+7.5*
All tokens (no pruning)	30.0 / 27.4	
Random tokens	28.3 / 27.4	
Random + PCD	36.8 / 38.2	
SemVID + PCD	34.3 / 34.4	
Coverage only	33.0 / 35.7	
BOCC (Coverage + PCD)	**39.7 / 43.3**	–

C. Checks against published numbers

Before comparing pruning methods under the TimeLens-Bench protocol, we reproduce the published unpruned scores with the TimeLens evaluation script, and check that our own inference pipeline gives the same scores on the same queries (Tab. 27).

D. End-snapping on S-EMBER

Setting every predicted end to the question time q raises every S-EMBER method to 42–46 mIoU, more than the 39.3 of BOCC, because 67% of the gold intervals end within the last fifth of the window. On closed intervals, end-snapping *hurts* every method (to 22–24 mIoU) while BOCC keeps 39.7; on QVHighlights it lowers BOCC from 48.8 to 40.5. A decoder that decides whether an interval is still open at question time is a natural extension. **[TODO: open-interval decoding.]**

Table 27. **Reproduction of TimeLens-Bench** (mIoU, no pruning). Top: TimeLens script on the full sets *vs.* published. Bottom: our pipeline *vs.* the TimeLens script on the same queries (paired Δ).

Model	Dataset (queries)	Ours	Reference	Δ
Qwen3-VL-8B	Charades-STA (3,363)	48.5	48.3 (published)	
Qwen3-VL-8B	QVHighlights (1,541)	59.6	59.4 (published)	
Qwen3-VL-8B	ActivityNet (4,500)	47.1	46.8 (published)	
TimeLens-8B	Charades-STA (3,363)	55.4	55.2 (published)	
TimeLens-8B	QVHighlights (1,541)	65.7	65.5 (published)	
TimeLens-8B	ActivityNet (4,500)	53.2	53.2 (published)	
Qwen3-VL-8B	Charades-STA (400)	48.0	47.5 (script)	+0.5
Qwen3-VL-8B	ActivityNet (400)	47.1	45.7 (script)	+1.4
Qwen3-VL-8B	QVHighlights (513)	60.5	60.5 (script)	0.0
TimeLens-8B	QVHighlights (513)	67.1	66.1 (script)	+1.0

E. Reproduction

BOCC corresponds to the following overrides of our code (released with the paper):

```
model=qwen3_vl_8b experiment=sember_grounding_uniform run.uniform_num_frames=64
dataset.grounding_prompt=timeline run.keep_time_tokens=surviving run.max_new_tokens=384
run.batch_size=8 run.prune_score=core_topk run.offline_keep_ratio=0.1
run.contrastive_mode=blind run.contrastive_alpha=1.0 run.contrastive_adaptive=true
run.contrastive_scope=time
(BOLT) run.retrieval=frames_cont run.retrieval_min_gap=1.0
```

Floors and paired intervals are computed by `docs/paper/analysis/floors.py`.